

DARSHIL MEHTA

Mumbai, India

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Education

Dwarkadas J. Sanghvi College of Engineering

2024 – 2028

B.Tech in Artificial Intelligence & Machine Learning (Honours in AI in FinTech) | CGPA: 9.1

Mumbai, India

Technical Skills

Languages: Python, SQL

Libraries/Frameworks: TensorFlow, NumPy, Pandas, Scikit-learn, FastAPI, Flask, LangChain, Sentence-Transformers

Tools & Platforms: Streamlit, Power BI, Git

Concepts: Machine Learning (Regression, Classification, Clustering), NLP, Deep Learning

Projects

PARIX – Student Intervention Analytics System | [GitHub](#) | [LiveTech Stack: Python, Streamlit, Scikit-learn, Pandas, GLM-4.](#)

- Built an AI-driven student risk analytics system on 6,000+ records using Linear Regression and K-Means clustering to identify at-risk students and behavioral personas.
- Automated intervention planning by integrating GLM-4.5 with a rule-based recommendation engine, generating personalized intervention and parent communication workflows.
- Deployed an interactive Streamlit dashboard enabling real-time student risk monitoring and intervention analysis.

Knowledge-Driven QA System (RAG-based) | [GitHub](#) [Tech Stack: Python, FAISS, Sentence-Transformers, Streamlit, Hugging](#)

- Built a RAG-based document QA system using BAAI/bge-m3 embeddings and FAISS, enabling sub-second semantic retrieval across 50+ documents.
- Reduced hallucinated responses by implementing similarity-threshold filtering and metadata-grounded retrieval with page-level citations.
- Developed an interactive Streamlit interface supporting context-aware document querying and source verification.

Laptop Price Prediction System | [GitHub](#) | [Live](#) [Tech Stack: Python, Scikit-learn, Pandas, Streamlit, XGBoost](#)

- Developed a Random Forest regression pipeline on 1,300+ laptop records, achieving R^2 score of 0.87 after hyperparameter tuning.
- Improved prediction performance through feature engineering, including PPI extraction, storage encoding, and target log transformation.
- Evaluated multiple ML models including XGBoost, SVR, KNN, and Linear Regression to optimize regression accuracy.
- Deployed a Streamlit web application for real-time laptop price estimation.

Resilient CRM – Offline-First Lead Management | [GitHub](#) [Tech Stack: Python, OCR, Regex, FastAPI, React, PostgreSQL](#)

- Developed an offline-first CRM system enabling reliable lead capture and synchronization in low-connectivity environments during a 24-hour hackathon.
- Built a regex-based OCR extraction pipeline to structure business card data and reduce manual lead entry effort.
- Improved data consistency before synchronization by implementing fuzzy-matching duplicate detection logic.
- Led OCR module development within a 3-member team and integrated offline-to-online background synchronization workflows.

Leadership & Extracurriculars

TechAI Mentee | *DJS Nova*

- Built and validated machine learning models utilizing large-scale astronomy datasets (SDSS) under senior mentorship.

Events Co-Committee Member | *DJS CodeAI*

- Coordinated logistics and operations for technical events, university hackathons, and developer workshops.